

DD_EM-CFG-04 - Approval Workflow Catalog

Version: v1.0

Status: Draft for review

Bundle: 10_other

Depends on:

- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- FOUNDATION_AUTH
- FOUNDATION_CAMUNDA
- FOUNDATION_KAFKA
- DD_EM-CFG-03-RBAC_Catalog-v1.0
- DD_EM-01-Franchise_Management-v1.0
- DD_EM-02-Contractor_and_Staff_Registry-v1.0
- DD_ICN-01-Internal_Communications-v1.0

1. Purpose

This DD defines the shared Approval Workflow Catalog for SOPHIX BSS.

Many modules need human approval:

- KYC L1/L2 approval;
- invoice adjustments, write-offs, credit notes, reversals;
- wallet manual adjustment;
- tax invoice cancellation;
- HomePass sensitive status transition;
- franchise activation/retirement;
- stock write-off and procurement approval;
- discount/campaign approval;
- dunning override or clear-without-payment;
- future CVM, field audit, and service-assurance compensation decisions.

EM-CFG-04 does not own those business objects. The owning module still owns the request, state, final apply/reject logic, and audit of its domain record.

EM-CFG-04 owns the reusable catalog that tells a module:

- whether approval is required;
- which approval workflow applies;
- which roles/permissions can approve;
- how many steps are needed;
- what thresholds and conditions apply;
- how user tasks should be routed;
- how approval outcome is reported back.

2. What This Module Is For

Use EM-CFG-04 when a module needs to decide:

Question	EM-CFG-04 answer
Does this action require approval?	Evaluate approval policy by module/action/context.
Which workflow template should run?	Return process key and step config.
Who can approve step 1/2/etc.?	Return approver role/permission/team scope rules.
Does amount/threshold change routing?	Apply configured condition rules.
Can requester approve their own request?	Enforce segregation-of-duties rules.
What should Backoffice show as a task?	Return user task metadata and allowed actions.
How should module receive final decision?	Module receives approved/rejected/revision outcome callback/event.

3. Scope

Included:

Capability	Purpose
Approval workflow catalog	Define reusable approval policies per module/action/operator.
Approval step catalog	Define ordered approver steps and required permissions.
Threshold and condition config	Amount, risk, status, channel, franchise, or reason-based routing.
Approval request registry	Track approval process instances started through this catalog.
Approval task/action APIs	Standard approve/reject/request-revision action contract.
Camunda template references	Associate policies with BPMN process keys.
RBAC integration	Use EM-CFG-03 roles, permissions, and scopes for approver eligibility.
Notifications/tasks	Emit internal task notifications through ICN.
Approval audit	Immutable approval decisions and actor comments.

Out of scope:

Out of scope	Owner
Business request creation	Owning module, for example BIL, ILM, RLM, OSR.
Final business apply/reject mutation	Owning module.
Authentication/JWT	FOUNDATION_AUTH.
Permission catalog	EM-CFG-03.
BPMN engine infrastructure	FOUNDATION_CAMUNDA.
Payment, invoice, customer, stock, HomePass domain records	Owning modules.
Complex risk scoring	Owning module or future risk/CVM service.

4. Ownership Pattern

EM-CFG-04 is a catalog and orchestration helper. It is not the owner of every approval request's business state.

Standard pattern:

1. Owning module receives a high-risk command.
2. Owning module validates the business request and creates its own pending request record.
3. Owning module calls EM-CFG-04 to evaluate approval policy.
4. If no approval is required, owning module applies request immediately.
5. If approval is required, EM-CFG-04 starts a configured approval process and creates approval tasks.
6. Approvers act through Backoffice/API.
7. EM-CFG-04 records decision audit and completes workflow step.
8. On final approved/rejected/revision outcome, EM-CFG-04 calls the owning module callback or emits an event.

9. Owning module performs final apply/reject/revision mutation.

Rule: EM-CFG-04 never directly updates invoices, wallets, customers, HomePasses, stock, tickets, or subscriptions.

5. Core Concepts

Concept	Meaning
Approval subject	The business object or request needing approval, for example adjustment request, KYC review, stock write-off.
Approval policy	Configuration that determines if approval is required and which process applies.
Approval step	One human decision point in the policy.
Approver rule	Role/permission/scope condition for who can act on a step.
Threshold	Numeric or categorical condition that changes routing.
Segregation of duties	Rule preventing requester and approver from being the same person or team.
Approval request	Runtime instance of an approval process started for one business subject.
Approval task	Runtime user task assigned to eligible approver(s).
Approval outcome	APPROVED, REJECTED, REVISION_REQUESTED, CANCELLED, EXPIRED.

6. Default Approval Use Cases

Use case	Owning module	Suggested policy
KYC L1 review	ILM/FUL-02	Single step: <code>customer.kyc.review_l1</code> .
KYC final approval	ILM/FUL-02	Second step: <code>customer.kyc.review_l2</code> .
Invoice adjustment low amount	BIL-02-ADJ	Auto-approve or one billing lead step.
Invoice adjustment high amount	BIL-02-ADJ	Billing lead then finance manager.
Invoice write-off	BIL-02-ADJ / BIL-04	Finance lead plus optional country head.
Wallet manual adjustment	BIL-05	Wallet admin plus billing lead.
Tax invoice cancellation	BIL-02-TAX	Tax admin approval.
HomePass transition into RFS/WAI/ACT	RLM-CFG-01 / FUL-07	HomePass admin or technical supervisor approval.
Franchise retirement	EM-01	Franchise admin plus operations lead.
Stock write-off	OSR-05 future	Inventory supervisor plus finance approval above threshold.
Procurement order	OSR-02 future	Procurement lead; finance approval above threshold.
Discount/campaign launch	SIP-03/SIP-05 future	Commercial lead plus finance/legal if high value.
Dunning clear without payment	BIL-04	Billing lead plus finance approval.

7. Data Ownership

EM-CFG-04 owns:

Entity	Purpose
<code>approval_policy</code>	Header definition for an approval policy.
<code>approval_policy_condition</code>	Conditions and thresholds that select workflow path.
<code>approval_step</code>	Ordered step definitions for a policy.
<code>approval_request</code>	Runtime approval instance linked to owning module subject.
<code>approval_task</code>	Runtime task visible to approvers.
<code>approval_decision</code>	Immutable approve/reject/revision action record.
<code>approval_callback_log</code>	Delivery attempts to owning module.

EM-CFG-04 references:

Reference	Owner
Permission/role codes	EM-CFG-03
User id	FOUNDATION_AUTH
Franchise scope	EM-01
Contractor/team scope	EM-02
Business subject id	Owning module
BPMN process key	FOUNDATION_CAMUNDA deployment

8. Core Tables

8.1 approval_policy

What it is for: defines one approval policy for a module action.

When to use it: a module asks which approval process applies to a command like write-off, KYC final approval, or franchise retirement.

Column	Type	Purpose	Sample
policy_id	TEXT PK	Policy id.	apol_bil_adj_wik_v1
operator_code	TEXT	Operator scope.	WIK
policy_code	TEXT	Stable business code.	BIL_ADJUSTMENT_APPROVAL
module_code	TEXT	Owning module.	BIL-02-ADJ
action_code	TEXT	Business action.	invoice.adjustment.approve_flow
display_name	TEXT	Admin label.	Invoice Adjustment Approval
status	TEXT	DRAFT, ACTIVE, RETIRED.	ACTIVE
approval_mode	TEXT	AUTO, SINGLE_STEP, MULTI_STEP, BPMN.	BPMN
camunda_process_key	TEXT NULL	BPMN process key if used.	approval_generic_threshold_v1
callback_url	TEXT	Owning module callback endpoint.	/internal/billing/adjustments/{subjectId}/approval-outcome
effective_from	TIMESTAMP	Start time.	2026-06-01T00:00:00Z
effective_to	TIMESTAMP NULL	End time.	null
created_at	TIMESTAMP	Creation time.	2026-05-29T12:00:00Z
updated_at	TIMESTAMP	Update time.	2026-05-29T12:00:00Z

Constraints:

- Unique active (operator_code, module_code, action_code).
- Active BPMN policy must have camunda_process_key.
- Active non-auto policy must have at least one active step.

8.2 approval_policy_condition

What it is for: stores rules that select auto/single/multi-step branches.

When to use it: amount thresholds, risk level, reason code, franchise, channel, or request type changes approval path.

Column	Type	Purpose	Sample
condition_id	TEXT PK	Condition id.	apc_adj_high_amount
policy_id	TEXT FK	Policy.	apol_bil_adj_wik_v1
condition_order	INT	Evaluation order.	20
field_path	TEXT	Input field to evaluate.	amount
operator	TEXT	LT, LTE, GT, GTE, EQ, IN, MATCHES.	GTE

Column	Type	Purpose	Sample
value_json	JSONB	Comparison value.	{"amount": 5000, "currency": "KES"}
result_mode	TEXT	Approval mode for this condition.	MULTI_STEP
result_step_set_code	TEXT NULL	Step set to use.	HIGH_VALUE
active	BOOLEAN	Whether condition applies.	true

Rule: conditions should be deterministic and simple. Complex eligibility/risk calculations belong to the owning module, which can pass `riskLevel` or `requiresApprovalReason` into the approval request.

8.3 approval_step

What it is for: defines one step in an approval policy.

When to use it: decide who can approve step 1, step 2, etc.

Column	Type	Purpose	Sample
step_id	TEXT PK	Step id.	aps_adj_finance_lead
policy_id	TEXT FK	Policy.	apol_bil_adj_wik_v1
step_set_code	TEXT	Branch/set this step belongs to.	HIGH_VALUE
step_order	INT	Execution order.	2
step_code	TEXT	Stable step code.	FINANCE_APPROVAL
display_name	TEXT	UI label.	Finance Approval
required_permission_code	TEXT	EM-CFG-03 permission needed.	billing.adjustment.approve
required_role_code	TEXT NULL	Optional role hint.	BILLING_LEAD
scope_type	TEXT	Scope required.	OPERATOR
assignment_strategy	TEXT	CANDIDATE_PERMISSION, CANDIDATE_ROLE, TEAM, EXPLICIT_USER.	CANDIDATE_PERMISSION
candidate_team_id	TEXT NULL	EM-02 team if used.	team_finance_approval
timeout_hours	INT	Escalation/expiry timer.	48
allow_requester	BOOLEAN	Whether requester can approve own request.	false
active	BOOLEAN	Whether step applies.	true

8.4 approval_request

What it is for: runtime approval process instance for one business subject.

When to use it: an owning module created a pending request and asked for approval.

Column	Type	Purpose	Sample
approval_request_id	TEXT PK	Approval request id.	apr_01J_ADJ_1001
operator_code	TEXT	Operator scope.	WIK
policy_id	TEXT FK	Policy used.	apol_bil_adj_wik_v1
module_code	TEXT	Owning module.	BIL-02-ADJ
subject_type	TEXT	Business subject type.	INVOICE_ADJUSTMENT
subject_id	TEXT	Owning module request id.	adj_01J_1001
requester_user_id	TEXT	User who requested action.	kc_user_billing_01
status	TEXT	PENDING, APPROVED, REJECTED, REVISION_REQUESTED, CANCELLED, EXPIRED.	PENDING
current_step_order	INT NULL	Current approval step.	1
camunda_process_instance_id	TEXT NULL	BPMN instance id.	cam_proc_01J_ADJ
context_json	JSONB	Approval facts/snapshot.	{"amount": 7000, "currency": "KES"}
created_at	TIMESTAMP	Creation time.	2026-05-29T12:30:00Z

Column	Type	Purpose	Sample
updated_at	TIMESTAMP	Update time.	2026-05-29T12:35:00Z

8.5 approval_task

What it is for: approver-facing task created from an approval step.

When to use it: Backoffice task inbox or module task queues.

Column	Type	Purpose	Sample
approval_task_id	TEXT PK	Task id.	apt_01J_ADJ_STEP1
approval_request_id	TEXT FK	Approval request.	apr_01J_ADJ_1001
step_id	TEXT FK	Step definition.	aps_adj_finance_lead
status	TEXT	OPEN, CLAIMED, COMPLETED, CANCELLED, EXPIRED.	OPEN
candidate_permission_code	TEXT	Permission that can act.	billing.adjustment.approve
candidate_role_code	TEXT NULL	Role hint.	BILLING_LEAD
candidate_team_id	TEXT NULL	Team if task routes to a team.	team_finance_approval
claimed_by_user_id	TEXT NULL	User who claimed task.	kc_user_finance_02
due_at	TIMESTAMP NULL	Due/timeout timestamp.	2026-05-31T12:30:00Z
created_at	TIMESTAMP	Creation time.	2026-05-29T12:30:00Z
updated_at	TIMESTAMP	Update time.	2026-05-29T12:31:00Z

8.6 approval_decision

What it is for: immutable record of each human decision.

When to use it: audit, dispute review, and debugging.

Column	Type	Purpose	Sample
decision_id	TEXT PK	Decision id.	apd_01J_APPROVE
approval_request_id	TEXT FK	Approval request.	apr_01J_ADJ_1001
approval_task_id	TEXT FK	Task.	apt_01J_ADJ_STEP1
decision	TEXT	APPROVE, REJECT, REQUEST_REVISION, CANCEL.	APPROVE
actor_user_id	TEXT	Approver user id.	kc_user_finance_02
comment	TEXT NULL	Approver comment.	Approved after invoice review.
decision_at	TIMESTAMP	Decision time.	2026-05-29T13:00:00Z
metadata_json	JSONB	Extra context.	{"ip": "10.1.2.3"}

8.7 approval_callback_log

What it is for: records callback/event delivery to owning module.

When to use it: retry failed outcome delivery and reconcile approval state.

Column	Type	Purpose	Sample
callback_log_id	TEXT PK	Log id.	acl_01J_ADJ_CB1
approval_request_id	TEXT FK	Approval request.	apr_01J_ADJ_1001
target_module_code	TEXT	Owning module.	BIL-02-ADJ
target_url	TEXT	Callback endpoint.	/internal/billing/adjustments/adj_01J_1001/approval-outcome
attempt_no	INT	Attempt number.	1
status	TEXT	PENDING, DELIVERED, FAILED_RETRYABLE, FAILED_FINAL.	DELIVERED
request_json	JSONB	Sent payload.	{"outcome": "APPROVED"}

Column	Type	Purpose	Sample
response_code	INT NULL	HTTP response code.	200
created_at	TIMESTAMP	Attempt time.	2026-05-29T13:05:00Z

9. APIs

All user-facing APIs require Authorization: Bearer <jwt>.

Write APIs require Idempotency-Key.

9.1 Policy Catalog APIs

API	Purpose	Consumers
POST /api/approval-policies	Create approval policy.	Backoffice admin, config import.
GET /api/approval-policies?operatorCode=WIK&moduleCode=BIL-02-ADJ	List policies.	Backoffice, modules.
GET /api/approval-policies/{policy_id}	Policy detail with conditions/steps.	Backoffice, modules.
PATCH /api/approval-policies/{policy_id}	Update draft policy.	Admin.
POST /api/approval-policies/{policy_id}/activate	Activate policy.	Admin/approval admin.
POST /api/approval-policies/{policy_id}/retire	Retire policy.	Admin.
POST /api/approval-policies/{policy_id}/conditions	Add condition.	Admin.
POST /api/approval-policies/{policy_id}/steps	Add step.	Admin.

9.2 Runtime APIs

API	Purpose	Consumers
POST /api/approvals/evaluate	Tell owning module whether approval is required.	Owning modules.
POST /api/approvals/requests	Start approval request.	Owning modules.
GET /api/approvals/requests/{approval_request_id}	Approval status/detail.	Owning modules, Backoffice.
GET /api/approvals/tasks?assigneeOrCandidate={user_id}&status=OPEN	User task list.	Backoffice.
POST /api/approvals/tasks/{approval_task_id}/claim	Claim task.	Approver.
POST /api/approvals/tasks/{approval_task_id}/approve	Approve task.	Approver.
POST /api/approvals/tasks/{approval_task_id}/reject	Reject task.	Approver.
POST /api/approvals/tasks/{approval_task_id}/request-revision	Ask requester to revise.	Approver.
POST /api/approvals/requests/{approval_request_id}/cancel	Cancel approval request.	Requester/module.

9.3 Evaluate Request Sample

```
POST /api/approvals/evaluate
Authorization: Bearer <service-token>
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "moduleCode": "BIL-02-ADJ",
  "actionCode": "invoice.adjustment.approve_flow",
  "requesterUserId": "kc_user_billing_01",
  "subjectType": "INVOICE_ADJUSTMENT",
  "subjectId": "adj_01J_1001",
  "context": {
```

```

    "amount": 7000,
    "currency": "KES",
    "reasonCode": "SERVICE_CREDIT",
    "franchiseId": "fr_01J_NRB_002"
  }
}

```

Response:

```

{
  "approvalRequired": true,
  "policyId": "apol_bil_adj_wik_v1",
  "approvalMode": "MULTI_STEP",
  "stepSetCode": "HIGH_VALUE",
  "camundaProcessKey": "approval_generic_threshold_v1",
  "steps": [
    {
      "stepOrder": 1,
      "stepCode": "BILLING_LEAD_APPROVAL",
      "requiredPermissionCode": "billing.adjustment.approve"
    },
    {
      "stepOrder": 2,
      "stepCode": "FINANCE_APPROVAL",
      "requiredPermissionCode": "billing.adjustment.approve"
    }
  ]
}

```

9.4 Start Approval Request Sample

POST /api/approvals/requests
 Authorization: Bearer <service-token>
 Idempotency-Key: approval-adj-01J-1001
 Content-Type: application/json

```

{
  "operatorCode": "WIK",
  "policyId": "apol_bil_adj_wik_v1",
  "moduleCode": "BIL-02-ADJ",
  "subjectType": "INVOICE_ADJUSTMENT",
  "subjectId": "adj_01J_1001",
  "requesterUserId": "kc_user_billing_01",
  "context": {
    "amount": 7000,
    "currency": "KES",
    "reasonCode": "SERVICE_CREDIT",
    "franchiseId": "fr_01J_NRB_002",
    "summary": "Credit customer for outage."
  }
}

```

Response:

```

{
  "approvalRequestId": "apr_01J_ADJ_1001",
  "status": "PENDING",
  "currentStepOrder": 1,
  "tasksCreated": 1
}

```

10. Business Rules

Rule	Description
APR-1	Owning module must create and validate its pending business request before starting approval.
APR-2	EM-CFG-04 must not directly mutate owning module business records.
APR-3	Approval policy selection is by operator, module, action, status, effective date, and conditions.
APR-4	Active non-auto policy must have at least one step.
APR-5	Approver eligibility is checked through EM-CFG-03 permission and scope.
APR-6	If <code>allow_requester = false</code> , requester cannot approve their own task.
APR-7	Every approval action must create immutable <code>approval_decision</code> .

Rule	Description
APR-8	Final outcome must be delivered to owning module through callback and/or event with retry.
APR-9	If callback fails, approval request remains in final approval state but callback log remains retryable until delivered or manually repaired.
APR-10	Retired policies remain readable for historical approval requests.
APR-11	Modules must support idempotent approval outcome callbacks.

11. Camunda And Worker Model

EM-CFG-04 can run simple single-step approvals without Camunda, but BPMN is recommended for:

- multi-step approvals;
- timer escalation;
- request revision loops;
- parallel approvals;
- operator-specific routing.

Reference BPMN service tasks:

Task/topic	Purpose
approval-load-request	Load approval request context.
approval-create-task	Create EM-CFG-04 task row and ICN notification.
approval-wait-user-action	Human user task.
approval-complete-step	Record step completion.
approval-deliver-outcome	Call owning module callback or emit final outcome event.
approval-expire-task	Mark timed-out task expired and escalate/reject per policy.

Worker names are logical topics. One worker deployment can handle multiple topics if metrics remain visible per topic.

12. Events

Event	Topic	Meaning	Consumers
ApprovalPolicyActivated	sophix.approval.policy-activated	Policy became active.	Backoffice, modules, audit.
ApprovalRequestStarted	sophix.approval.request-started	Runtime approval started.	Backoffice, ICN, REP.
ApprovalTaskCreated	sophix.approval.task-created	Human task created.	ICN, Backoffice task inbox.
ApprovalTaskCompleted	sophix.approval.task-completed	Task approved/rejected/revisioned.	REP, audit.
ApprovalRequestApproved	sophix.approval.request-approved	Final approved outcome.	Owning module, REP.
ApprovalRequestRejected	sophix.approval.request-rejected	Final rejected outcome.	Owning module, REP.
ApprovalRevisionRequested	sophix.approval.revision-requested	Revision requested.	Owning module, requester UI.
ApprovalRequestExpired	sophix.approval.request-expired	Request timed out.	Owning module, Backoffice.

13. Frontend Usage

13.1 FE-APP-01 Backoffice

Backoffice uses EM-CFG-04 for:

Area	API
My approval tasks	GET /api/approvals/tasks?assigneeOrCandidate={user_id}&status=OPEN
Approval detail	GET /api/approvals/requests/{approval_request_id}
Approve/reject/revision	task action APIs
Approval policy admin	policy catalog APIs
Pending approval badges	owning module detail plus approval request status

UI rule: Backoffice displays the business subject summary, but detailed fields should come from the owning module. Approval UI should not reconstruct invoices, customers, or stock records from approval context alone.

13.2 Owing Module Screens

Each module screen should show:

- pending approval status;
- current approval step;
- who can act;
- approval history;
- final outcome and callback status if failed.

14. Integration With Other Modules

Module	Integration
EM-CFG-03	Checks approver permission/scope.
ICN-01	Sends internal approval task notifications.
BIL modules	Start approvals for adjustments, write-offs, wallet operations, tax cancellation, dunning overrides.
ILM/FUL	Start approvals for KYC and customer-sensitive changes.
RLM/FUL-07	Start approvals for HomePass sensitive transitions.
EM-01	Start approvals for franchise retirement/suspension if configured.
OSR future	Start approvals for procurement and write-off.
SIP future	Start approvals for campaigns, discounts, bundle launches.
REP-01	Consumes approval metrics and audit events.

15. Infrastructure And Deployment

For a budget-conscious medium deployment:

Component	Recommendation
Runtime	Run inside EM/business-services or workflow-services pod.
Database	PostgreSQL em or approval schema in shared business DB.
Camunda	Use existing Camunda cluster from foundation.
Workers	Consolidate approval worker topics with workflow worker deployment.
Kafka	Use transactional outbox for approval events.
Cache	Cache active policy lookup briefly, 2-5 minutes. Invalidate on policy activation/retirement.
Notifications	ICN event or internal API for task notifications.
Observability	Alert on overdue tasks, callback failures, stuck BPMN instances, outbox backlog.

AWS recommendation:

Concern	AWS guidance
Compute	Shared ECS/EKS services for approval API/workers.
DB	Existing RDS PostgreSQL.
Camunda	Existing Camunda deployment/database.
Events	Existing MSK/Kafka.
Secrets	Secrets Manager/SSM for service tokens/callback credentials.
Monitoring	CloudWatch/Prometheus alerts for pending tasks and callback retries.

16. Tests

Minimum tests:

Test	Expected result
Evaluate action with no matching policy	Returns approval not required or configured default.
Evaluate amount below auto threshold	Returns <code>approvalRequired = false</code> if policy allows auto.
Evaluate high amount	Returns multi-step policy/steps.
Start approval request	Request and first task created.
Requester attempts to approve own task where not allowed	403 or task action denied.
User without required permission attempts approval	403.
Approve final step	Final outcome delivered to owning module.
Reject step	Final rejected outcome delivered to owning module.
Request revision	Revision outcome delivered, request no longer open.
Callback failure	Callback log retryable; approval state preserved.
Duplicate approval request idempotency key	Same response, no duplicate request.
Retired policy remains readable for historical request	Historical detail still works.

17. Open Decisions For Review

Decision	Recommendation
Should all approvals use Camunda?	No. Single-step/simple approvals can be DB-driven. Use Camunda for multi-step/timer/revision flows.
Should EM-CFG-04 own approval tasks for all modules?	Yes for shared task visibility, but owning module still owns business state.
Should approval decisions be delivered by callback or Kafka event?	Support both; callback is immediate, event is useful for async/reconciliation. Owning modules must be idempotent.
Should approver lists be users or permissions?	Prefer permission/scope-based candidate resolution. Use explicit users only for exceptions.
Should requester self-approval ever be allowed?	Default no. Allow only for low-risk policy with explicit <code>allow_requester = true</code> .